SkinAid: A GAN-based Automatic Skin Lesion Monitoring Method for IoMT Frameworks

Prathistith Raj Medi[1], Praneeth Nemani[2], Pitta Vivek Reddy[3], Venkanna Udatalapally[4][0000−0002−3596−6679], Debanjan Das[5][0000−0002−3450−9767], and Saraju P. Mohanty[6][0000−0003−2959−6541]

[1] Department of Data Science and Artificial Intelligence, IIIT Naya Raipur, India.
prathistith19102@iiitnr.edu.in
[2] Department of Computer Science and Engineering, IIIT Naya Raipur, India.
praneeth19100@iiitnr.edu.in
[3] Department of Computer Science and Engineering, IIIT Naya Raipur, India.
pitta19100@iiitnr.edu.in
[4] Department of Computer Science and Engineering, IIIT Naya Raipur, India.
venkannau@iiitnr.edu.in
[5] Department of Electronics and Communication Engineering, IIIT Naya Raipur, India.
debanjan@iiitnr.edu.in
[6] Department of Computer Science and Engineering, University of North Texas, USA.
smohanty@ieee.org

Abstract. Not knowing the type of skin lesions or too much late in diagnosis can lead to chronic disease or skin cancer. In SkinAid, we propose a system that can not only monitor and detect skin disease but also create awareness for the user to care for the skin. SkinAid provides information on the skin lesions along with the classification of skin lesions to Malignant-Skin or Normal-Skin. In this regard, the highly unbalanced and limited data leads existing Deep Learning solutions to overfit or poorly generalize resulting in reduced performance. We explore Generative Adversarial Networks (GANs) to enhance the dataset. Also, we proposed a Deep Convolutional Neural Network (CNN) model for edge computing platforms which can automatically detect, classify skin lesions from the captured image of the user's skin. A single board computer and a smartphone used as edge platforms have been implemented in SkinAid. The SkinAid model has produced an overall accuracy of 92.2% and classifies 7 different skin lesions, and also detects if the diagnosed lesion is Malignant or Benign. The patient can capture the real-time skin lesion images using a smartphone app or camera and diagnose the skin lesions in Internet-of-Medical-Things (IoMT) platform for remote monitoring by dermatologists.

Keywords: Smart Healthcare, Internet of Medical Things (IoMT), Generative Adversarial Networks (GAN), Convolutional Neural Networks (CNN), Skin Health, Skin Cancer, Deep Learning, Image Contours.

1 Introduction

The term "lesion" refers to any part of the body that is abnormal or dissimilar to other parts. Similarly, the term skin lesion refers to those patches of skin that appear deviant

from the rest of the skin. Some visible examples of deviants include irregular coloring of skin and itching. Skin Lesions are classified into many classes based upon different characterizations. Based upon their origin and composition, they are classified into three types: primary and secondary Lesions. Primary lesions refer to those lesions which are inherited at childbirth or gradually developed over time owing to genetics, lifestyle, and other factors. Secondary lesions refer to those lesions that develop over primary lesions when exposed to lesion causing agents over some time. Some examples of primary lesions include blisters, rashes, and nodules. Examples of secondary lesions include Actinic Keratosis, vascular Lesions, etc. Skin lesions are classified into cancerous and non-cancerous Lesions based upon their ability to be transformed into Skin Cancer [20]. Skin cancer is the condition of the abnormal development of skin cells. The other terms for the above categorization are melanomic and non-melanomic skin lesions respectively [5]. Some example of cancerous lesions includes melanoma [4] [18] and basil cell carcinoma while non-cancerous lesions include melanocytic nevi, actinic keratosis, etc.

However, many lesions go unidentified or unrecorded due to many factors [12]. Statistics show that number of people diagnosed with cancer-causing skin lesions are estimated to be 9500 per day in the United States. The average cost of treating skin cancers in the United States is, projected to be 8.1 billion dollars, with non-melanoma skin cancers costing 4.8 billion dollars and melanoma skin cancers costing 3.3 billion dollars. Studies by dermatologists depict that a majority of skin lesions are harmless. But there is a need for differentiation of these lesions from malignancy. According to a survey by the Skin Cancer Foundation, 99% of skin cancers can be cured by the early detection of skin lesions.

In the new age of learning, Deep Learning and Computer Vision frameworks provide the best possible solutions to tackle this problem. But one of the major shortcomings of these frameworks is the deficiency of data. Due to the high number of skin lesions and their different variants in various people, there is always a deficiency of skin data. ISIC, HAM10000, PAD-UFES are some of the major datasets of skin lesions. Though they are claimed to be state-of-the-art datasets, they still lack data about many classes of lesions and their variants. In recent years, CNNs [2] were proven to produce the most accurate and precise results for developing skin-lesion classifiers. The main objective of the usage of deep neural networks over shallow neural networks is that when more layers are present, the intermediary features which are present between the raw data and the final output are very effectively learned. The network weights are modified using partial derivatives derived from the previous layer's error function. The function of updates in the weights is nullified when these partial derivatives start to become very small.

Non-cancerous Lesions like actinic keratosis, melanocytic nevi, benign keratosis, vascular lesions, dermatofibroma and cancerous Lesion like melanoma [7] and basil cell carcinoma are the seven types of skin lesions detected and classified through this research [16]. A Wasserstein GAN is used to generate synthetic data from the original data, and a CNN model is trained to detect and classify skin lesions. Our concept revolves around developing a mobile application that takes a skin lesion image as input

and performs the classification of the image into the respective class using a trained CNN model.

This work proposes "SkinAid", an IoMT-enabled mobile application that could detect and classify 7 different skin lesions from images captured with a Smartphone. The motivation behind SkinAid is to provide the following:

1. Consistent detection and monitoring.
2. Immediate access of medical support without internet connectivity disregarding the location.
3. An approach that signifies Early detection can prevent severe fatality.

The Conceptual overview of SkinAid where the skin lesion image is captured in the SkinAid mobile app and the user is notified about the class of lesion is shown in Fig. 1.

Fig. 1: Conceptual Overview of SkinAid

The paper is organized in the following sections: Section 2 gives the overview of prior related work. The Novel contributions of the proposed system is discussed in Section 3. Section 4 details the proposed methodology followed in making of SkinAid. Further, Section 5 discusses specifications used in the proposed solutions and the experimental results. Finally, Section 6 concludes the paper.

2 Related Research Overview

The significance of IoMT Smart healthcare is that there is effective diagnosis of different potential diseases, proper treatment of these diseases and improvement in the qual-

ity of life. Currently, research in IoMT Smart healthcare is seen as a multi-dimensional scenario where many parameters are taken in account. Some examples of research in this field include human body monitoring, stress monitoring and Food in-take monitoring [15] [14].

Smart skin lesion detection and classification has been a topic of interest for a few years now. Numerous different implementations of skin lesion detection have been attempted. But some of these efforts are limited to binary classification, i.e., benign (non-cancerous) or malignant (cancerous), and data augmentation techniques were also absent, with the sole use of the under-sampling technique, resulting in over-fitting and poor generalization [17]. A method combining deep CNNs and traditional computer vision features based on clinical criteria with unbalanced and limited data is proposed in [9]. However, the model achieved a maximum training accuracy of 85.5%.

Fusing the deep features from various pre-trained CNNs is shown to lead to better classification performance in [11]. But it was restrictive to only 3 skin lesions and included only AlexNet, VGG-16, and ResNet-18 as their pre-trained models while there were many state-of-the-art CNN architectures. A new prediction model based on a new regularizer was presented in [1], which achieved high accuracy. However, it was limited to binary classification, and choosing the regularization parameter was difficult as mentioned by the author.

Although extensive research has been conducted in this field, to the best of our knowledge, none of these solutions has a preprocessing pipeline with cropping contours, global contrast normalization and morphological transformations. None of solutions used advanced data augmentation techniques like GANs for generating synthetic data. Most of them do not have a dedicated user interface like a smartphone application build with the trained deep learning model for practical real-world use.

The SkinAid not only ensures that it is detecting the class of skin lesion captured from the smartphone camera but also predicts if the lesion is malignant(cancerous) or benign(non-cancerous). Additionally, it also generates a medical report and provides information about the diagnosed lesion, including the contact details of the nearest available dermatologist for remote monitoring. The following Table 1 shows the comparison of SkinAid with existing solutions of skin lesion classification.

3 Novel Contributions

The identification and classification of the different kinds of skin lesions have always been a matter of contention among different researchers. Many solutions were proposed and some of the disadvantages included less data, over-fitting on the training set, improper preprocessing, binary classification, traditional CNN architectures, and absence of a novel application, etc. To address the above disadvantages, we have proposed “SkinAid” The significance of the proposed system include the following:

Table 1: Comparison of SkinAid with exisiting works

Name of the Work	**Methodology Used**	**Dataset**	**Accuracy**	**Interface Present?**	**Over-fitting?**	**No. of Lesions Classi-fied**
Skin Lesion Classification from Dermoscopic Images [17]	VGGNet	Dermofit Image Library	81.3%	No	No	2
Deep Learning Feature Combining Classification [9]	Modified ResNet50	ISIC 2018	85.5%	No	No	7
Skin Lesion Classification using Hybrid NNs [11]	SVM Classifiers	ISIC 2017	83.3%	No	Partial Over-fitting	2
Skin Lesion Classification Using Convolutional Neural Network With Novel Regularizer [1]	Novel Regularizer	ISIC Archives	97.2%	No	No	2
SkinAid	**WGAN-GP with DenseNet-121**	**HAM-10000**	**92.2%**	**Yes**	**No**	**7**

- The presence of state-of-the-art image preprocessing techniques to enhance the process of feature extraction
- The generation of synthetic data from the existing data by the usage of Wasserstain GAN.
- The training of the augmented data upon a CNN architecture thus obtaining a higher accuracy as compared to traditional Deep Learning Techniques.
- A robust android application is developed which can predict the class of the skin lesion on real-time data.

4 Proposed Novel SkinAid in an IoMT Framework

SkinAid provides the patient with a mobile app that can be used to detect and classify skin lesions by exploiting the computation power and digital lens of smartphones. This forms the next wave of advancement in smart skincare. The methodology is organized in the following fragments:

1. **Data preprocessing:** The data is pre-processed to remove unnecessary noise and features which might lead to misclassification of the lesions. The techniques used for preprocessing are contour Detection, GCN, and morphological Transformations.
2. **Generation of new samples:** From the existing data, new synthetic data is generated with the help of W-GAN and the data augmentation is performed. This enhances the data to train upon for the CNN Model.

3. **CNN Training:** The state-of-the-art CNN model is trained, tested, and validated on the augmented data. The accuracy is recorded and the model is converted into a ".pb" file for android deployment.
4. **Deployment:** The trained weights are deployed in android with the help of JAVA. A user interface is created which detects and classifies the different lesions and provides suggestions of a dermatologist.

The outline of the proposed approach is shown in Fig. 2 and subsequently, it is explained in detail in the following sub-sections.

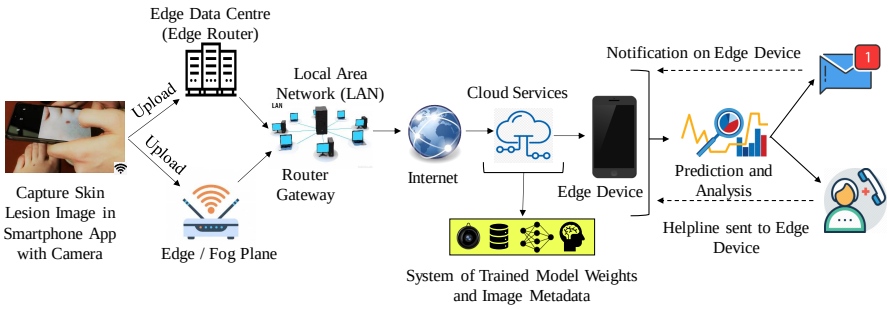


Fig. 2: Proposed IoMT based H-CPS for Smart Healthcare with SkinAid.

4.1 Dataset

Dataset plays an important role in the entire application. The accuracy and performance of our CNN model dramatically depend on the quality of input image data. Recently, few skin research organizations made their skin lesion data available publically. Here, in this paper, we use the HAM10000 dataset, which is a collection of various skin lesions and dermatoscopic images. The HAM10000 dataset consists of 10,015 skin lesion images with 7 different classes of lesions, and this dataset is publically accessible [19]. Unfortunately, it is limited and has highly unbalanced classes, and if directly trained on CNN architectures, it would over-fit and fail to generalize, leading to poor performance. Traditional data augmentation techniques like under-sampling and oversampling did not increase the performance significantly. To address this problem, we pre-process the entire dataset and split it into training and validation sets. Then we use the training set to train the W-GAN and generate synthetic data, thereby increasing our training data and overcoming the highly unbalanced classes and limited data problem. Finally, we train on different CNN architectures, validate with the validation set, and deploy our best model in the Smartphone application.

4.2 Image preprocessing

The images present in the dataset are not uniformly arranged. There are disparities among the lesions of the same class and different classes. There is also the presence of

noise and unnecessary background which leads to the presence of undesired features. The significance of image preprocessing is that it reduces the noise and other factors leading to misclassification to a great extent.

The preprocessing of images consists of 3 steps as shown in the flow diagram (Fig. 3)

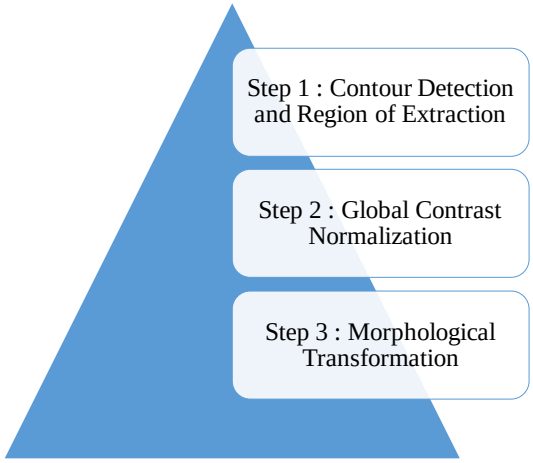


Fig. 3: Image preprocessing Overview

1. **Contours:** In a broader perspective, contours are referred to as a drawn outline of an irregular figure. In our context, they refer to the outlines of the skin lesion. Here we use contours, for detecting lesion images and extracting the region-of-interest (ROI), including only the skin lesion, thereby excluding the unwanted background. This reduced a lot of time taken by manual cropping. We aim to determine the bounding box of the lesion in its minimum enclosing rectangle. The $cv2.findContours$ function in conjunction with another few OpenCV utilities makes this very easy to accomplish. Further, all the images were resized to $(224 \times 224 \times 3)$ since they had different dimensions. Algorithm 1 depicts the above process.
2. **Global Contrast Normalization:** Contrast is different from another thing. In image processing, it is referred to as the luminance difference between bright and dark pixels that makes an object distinguishable in an image. Here, global contrast normalization is essential because images can have varied contrasts and if trained on these images our model might perform poorly. Some images can have high contrast and some can have low contrast, which results in losing important features and having compact information in images respectively. Therefore, we apply global contrast normalization by calculating the mean of an image and subtracting it from each

Algorithm 1 Contour Detection and Extraction of ROI

Function Contour Detection $(image)$
Input: Dataset Image
Output: Contour Detected Image
if ***Image*** *is* $\neq$ ***None*** **then**
  *img : cv2.**imread**(image path)*
  *grey : img.**cvtColor**(RGB to Grey)*
  *Thresh-Image : cv2.**threshold**(grey)*
  *Contours : cv2.**findContours**(Thresh-Image)*
  return *Contours;*
else
  return *None;*
end

pixel, and then divide each pixel by the standard deviation. All these calculations and operations are implemented using NumPy in python. This helps in improving the performance of our model with better feature extraction as shown in Algorithm 2. For an image (i.e. 3D tensor) $X \in \mathbb{R}^{r \times c \times 3}$, wherein $X_{i,j,k} \in \{1, 2, 3\}$ repre-

Algorithm 2 Global Contrast Normalization

Function Global Contrast Normalization $(Contours, \lambda, \epsilon)$
Input: Contours, λ, ϵ
Output: $GCNImage$
if ***Contours*** *is* $\neq$ ***None*** **then**
  img : cv2.imread(Contours)
  x : **np.array**(img)
  Average$_x$: **np.mean**(x)
  $x : x - Average_x$
  $contrast : \sqrt{(\lambda + \textbf{np.mean}(x^2))}$
  $GCN : s * x / \textbf{max}(contrast, \epsilon)$
  return GCN
else
  return None;
end

senting the weights of Red, Blue and Green (RGB) colour channels respectively at positions i, j and k. More precisely, GCN produces output image $\overline{X}$.

$$X^{|} = s \frac{X_i, j, k - \overline{X}}{\epsilon, \sqrt{\lambda + \frac{1}{3r} \sum_{i=1}^{r} \sum_{c}^{j=1} \sum_{3}^{k=1} (Xi, j, k - \overline{X})^2}} \tag{1}$$

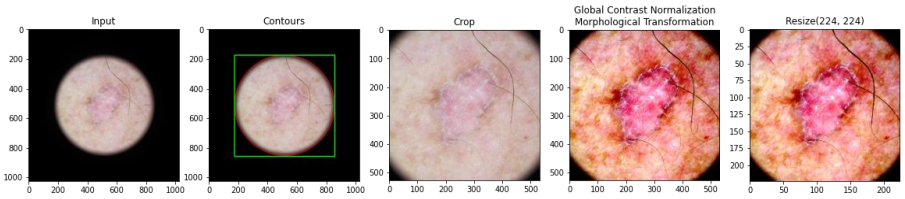
3. **Morphological Transformation:** Generally, while capturing images, there is always a chance that image pixels could be spatially disconnected, though it makes

Algorithm 3 Morphological Transformations

Function Morphological Transformation (GCN)
Input: GCN
Output: Image$_{Final}$
if ***GCN*** *is* $\neq$ ***None*** **then**
  img : cv2.imread(GCN)
  x : **np.array**(GCN)
  kernel : **np.array**(1,1)
  Image$_{Final}$: cv2.morphologyEx(image, kernel)
  return $Image_{Final}$
else
  return None;
end

a minor fragment of the image to be noisy. This problem can be solved using morphological transformation [10]. Here two functions of morphological transformation are useful they are dilation and erosion. Dilation helps in connecting adjacent pixels by turning on the pixel value, and erosion does the reverse. Thus, using the dual of dilation and erosion simultaneously is considered an efficient technique for noise reduction. The complete dataset consisting of $10,015$ images with a kernel of $1x1$ has been morphologically transformed with dilation followed by erosion as shown in Algorithm 3.

The outcome of image preprocessing is depicted in Fig. 4. In order to perform the above task, a sample image from the dataset is taken which is shown in Fig. 4a. Subsequently, Figures 4b, 4c and 4d show the preprocessing steps. Finally, Fig. 4e shows the results of image resizing.



(a) Sample image from HAM10000 dataset

(b) Contour detection with OpenCV to extract Region-of interest (ROI)

(c) Cropping biggest square contour

(d) Applying GCN and Morphological Transformation on extracted ROI

(e) Resizing all images to224 x 224 pixels

Fig. 4: Image preprocessing Results

4.3 Splitting the Dataset:

We split our dataset into two parts: the training set and the validation/test set. The validation set includes 10% of samples, whereas the training set includes 90% of samples

from the original dataset. The training examples are used to train the model while training, backpropagation finds the optimal weights for the model. The validation split of our data is used to fine-tune the model's parameters being trained and also for testing the trained model as it contains images of all different skin lesions and analyzes the trained model's performance. The following Table 2 shows the splitting statistics of the data.

Table 2: Splitting of the dataset

Partition Type	Number of Samples
Train	8912
Validation/Test	1098
Number of Samples	10015

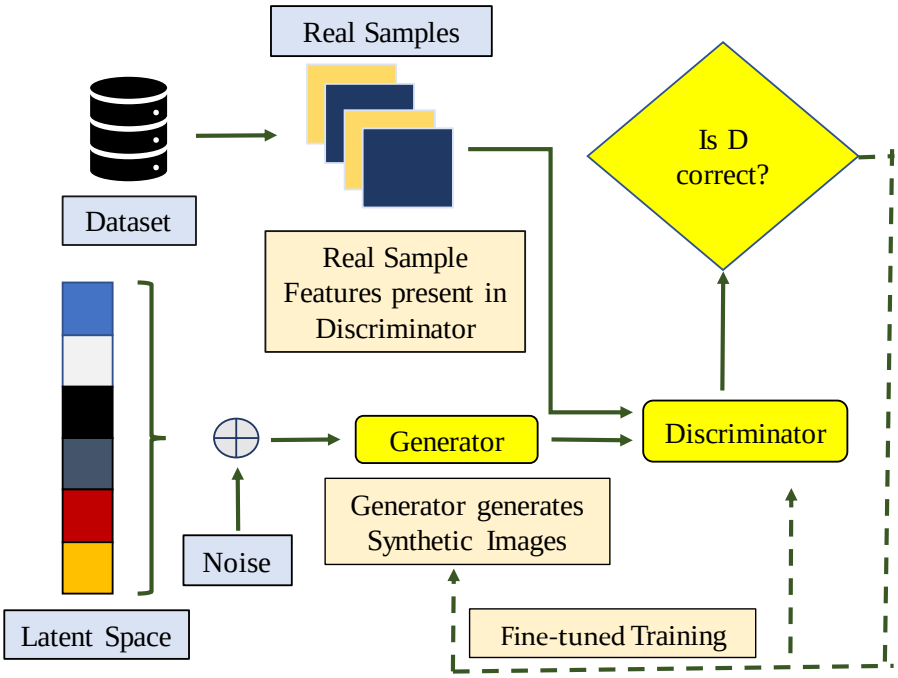


Fig. 5: GAN Architecture to generate Synthetic images of skin lesions.

4.4 Data Augmentation with WGAN:

Due to the highly unbalanced classes and limited data of original skin lesion images in our dataset, it is prone to over-fitting and poor performance if directly trained on CNNs. Therefore, we decided to use data augmentation techniques and generate synthetic data with GANs to overcome the above challenges [13]. Fig. 5 shows the proposed GAN Architecture.

For training the generative adversarial networks (GANs), we used the pre-processed training set images. GANs are designed with two parts they are: Generator (G) and Discriminator (D). Both of these structures are nothing but CNNs, and they compete with each other in an adversarial way. After training for few epochs, GANs can generate synthetic samples that are similar/identical to the original data. They have great potential as they can generate new identical images and can be trained to emulate any data patterns or distributions. Z is the input to G, which is a random noise vector generally picked from a Gaussian distribution. Using this input G generates an image, and the D tries to classify or distinguish if the output image from the G is original (real image) or fake (generated image). Upon training, G learns to mimic the original data and generate images close to real images, thereby fooling the discriminator. G learns by inspecting and analyzing every step and updating its parameters. Below is the mathematical expression (Equation 2) showing the relation between G and D,

$$G = \min_{G}\max_{D}[\mathbb{E}_{x\sim\mathbb{P}_r}[log(D(x))] + \mathbb{E}_{\widetilde{x}\sim\mathbb{P}_g}[log(1 - D(\widetilde{x}))]]. \tag{2}$$

However, due to frequent mode collapses and often failing to converge, training basic GAN is challenging. Many different approaches have been attempted to overcome these challenges. In this paper, we use the Wasserstein GAN with gradient penalty to generate synthetic data [3]. This special GAN uses gradient penalty instead of weight clipping which was used in the prior version of WGAN to impose on the Lipschitz constraint, which is defined as a function $f(t, y)$ satisfy 1-Lipschitz constraint for variable y on a set D $\in$R^2 if there exists L such that $L > 0$ and

$$L = |f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|. \tag{3}$$

Where L is Lipschitz constant.In WGAN, discriminator (D) has a unique purpose and is called a critic. Its primary function is to classify or distinguish if the image is original (real image) or fake (generated image). Additionally, the use of gradient penalty simulates the Lipschitz function more tightly, therefore ensuring the accurate calculation of Wasserstein distance [6]. The new equation (Equation 4) with gradient penalty showing the relation between generator (G) and discriminator (D) is depicted below

$$L = \mathbb{E}_{\widetilde{x}\sim\mathbb{P}_g} log(D(\sim x)) - \mathbb{E}_{x\sim\mathbb{P}_r} log(D(x)) + \lambda E_{x\sim\mathbb{P}_r}[\nabla_{\hat{x}}||(D(\hat{x})||_2 - 1)^2]. \tag{4}$$

In our approach, we constructed both Critic and Generator with five convolution layers with activation function as Leaky ReLU with a negative slope of 0.2, including six Convolution Transpose blocks and Leaky ReLU activation function of critic and batch and normalization respectively. We have resized the training set to $224 \times 224 \times 3$

Table 3: Augmented dataset

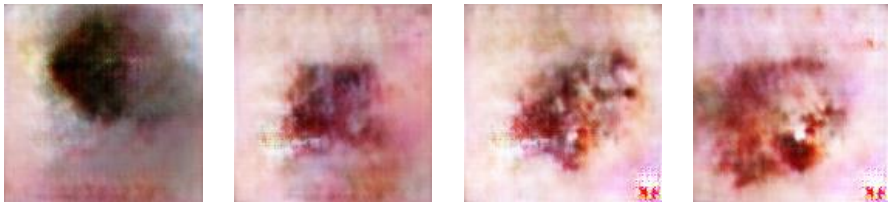
Lesion Type	Original Data	Train	Validation/Test	Augmented Final Dataset
Melanocytic Nevi	6705	5822	883	5822
Melanoma	1113	1067	88	5335
Benign Keratosis	1099	1011	46	5055
Basal Cell Carcinoma	514	479	35	4790
Actinic Keratosis	327	297	30	4455
Vascular Lesions	142	129	13	5160
Dermatofibroma	115	107	8	5350
Total	10015	8912	1098	35967

and fed it to the network for training. The optimizer used here is Adam with a learning rate, α = $10^{\neg 4}$, value of β = (0, 0.9) and the value of n_{critic} = 5. A total of 27,055 new synthetic images were generated of dimensions $224 \times 224 \times 3$ and were added to the original dataset. Table 3 depicts the metadata of the augmented dataset.

A total of 27,055 new synthetic images were generated of dimensions $224 \times 224 \times 3$ and were added to the original dataset. Let I_{train}, $I_{validation}$ denote the training and validation sets of the image data before the training of the GAN. I_{gen} denotes the data generated by the GAN. The new training and validation sets are denoted by the following equation:

$$I_{train}, I_{validation} \equiv (I_{train} + I_{gen}), (I_{validation} + I_{gen}). \tag{5}$$

The below Fig. 6 shows a sample of the synthetic data generated by the W-GAN depicting each different lesion class. Fig. 6a shows a sample of synthetic data generated for the class Actinic Keratosis. Similarly, Figures 6b, 6c and 6d show the samples of the synthetic data generated for the classes Melanoma, Melanocytic Nevi and Vascular Lesions respectively.



(a) Sample 1 : Actinic Keratosis (b) Sample 2 : Melanoma (c) Sample 3 : Melanocytic Nevi (d) Sample 4 : Vascular Lesions

Fig. 6: Sample of Synthetic Data generated by W-GAN

4.5 Training CNN model:

The primary goal is to train a deep learning model that can detect and classify different classes of skin lesions with high accuracy and generalization. Here, we use a CNN model. The main advantage of using CNNs is its potential to learn and extract the best and relevant features for better image classification, the equation is represented below:

$$G[m,n] \equiv (f * h)[m,n] \equiv \sum_j \sum_k h[j,k] f[m-j, n-k]. \tag{6}$$

The input image is represented as f, and h represents the filter or kernel in the equation as defined above. Here m and n represent the indexes of rows and columns of the result matrix, respectively. A pre-defined CNN architecture is used, which acts as a checkpoint for training the skin lesion prediction model. Many pre-trained CNN architectures are available, such as DenseNet-121, ResNet-50, VGG-16, AlexNet [8], etc. Here, we use Transfer Learning with different CNN architectures; using this is advantageous as it significantly reduces the training time and gives us better performance even with smaller datasets. Therefore, we use pre-trained CNN architectures like DenseNet-121, ResNet-50, VGG-16, AlexNet, etc., with Imagenet weights. Using the existing pre-trained model as a base on top of dense layers is advantageous as it boosts learning by adding to the existing knowledge, thus giving better classification results. The image classifier is trained on top of pre-trained base models like DenseNet-121, ResNet-50, VGG-16, etc., to find the best architecture with the best performance. Out of the above architectures, DenseNet-121 has achieved the maximum accuracy of 92.2%.

4.6 Tensorflow Serving

The trained weights generated after training the model are saved using the .save function of Pytorch. The trained weights are then used for deployment purposes in the form of a saved ".pth" file. To develop a industry-grade prototype, the most efficient procedure is to convert the ".pth" file extension into a ".pb" file extension. To achieve this conversion, we use ONNX, which can convert a .pth file to a .pb file. Finally, using the TensorFlow Serving, we can create an industry-grade app by embedding the trained TensorFlow model.

4.7 Deployment

The user interface consists of two steps. In the first step, the user identifies his symptoms from the given options in the interface. These symptoms are recorded for further analysis. In the second step which is the Image Analysis step, the user uploads an image of the skin lesion. The model converts the given image into $224X224$ dimensions and predicts the class of the disease. The class of the disease is displayed on the screen along with the symptoms recorded. Amidst the 7 classes, the classes of Melanoma and Basil cell carcinoma are two kinds of skin cancer Lesions. The remaining classes are not cancer lesions. The above notification is also displayed on the screen. The following Fig. 7 shows the methodology of deployment and the creation of the user interface.

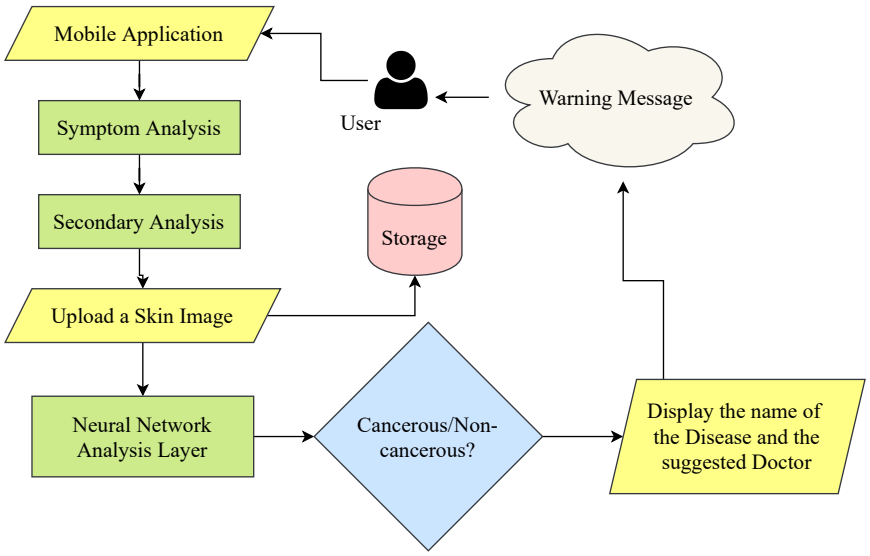


Fig. 7: Development of user warning in SkinAid

5 Results and Experimental Analysis

The following section emphasizes on the validation, testing and performance of the SkinAid application along with the hardware specifications required for W-GAN and SkinAid

5.1 Hardware specifications of the SkinAid App

The SkinAid app was tested on the conditions of 4GB RAM, 12GB free internal memory and 48MP android camera. The minimum conditions for the efficient performance of SkinAid are 2.5GB RAM, 150Mb storage and 8MP android camera. The below Table 4 shows the hardware specifications of the SkinAid app.

Table 4: Hardware specifications of SkinAid

Hardware specification	**Testing Conditions**	**Minimum Conditions**
RAM	4GB	2.5GB
Internal Memory	12GB	150MB
Camera	48MP	8MP
Internet Connectivity	50Mbps	Nil

5.2 Hardware Specifications for W-GAN

The W-GAN was trained on the Google Colaboratory GPU whose specification maps to Nvidia T4 whose GPU memory clock frequency was 1.59GHz. The available RAM was about 12GB and the pre-processed dataset size was about 2GB. The data was trained upon 2000 epochs with the internet bandwidth being equal to 50Mbps. The following Table 5 shows the hardware specifications of the trained GAN model.

Table 5: Hardware specifications for training of the W-GAN

Hardware specification	**Testing Conditions**
GPU	Nvidia T4
Available RAM	12GB
Initial Dataset Size	2GB
GPU Memory Clock	1.59 GHz
Internet Connectivity	50Mbps

5.3 User Interface and Android Application Results

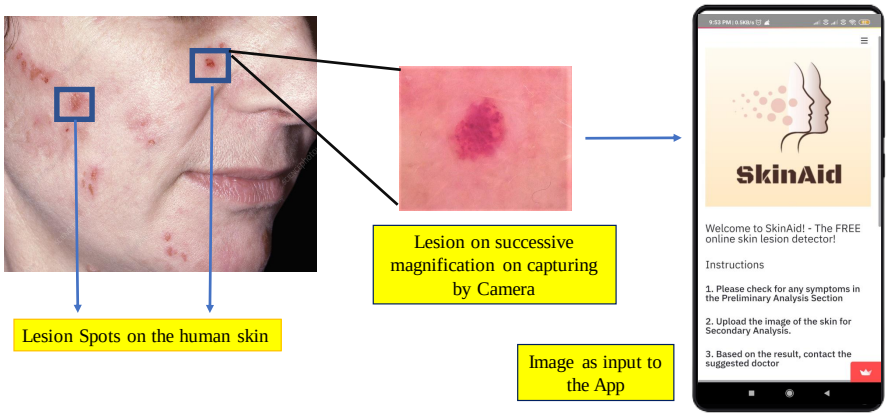
The user first identifies the lesion on his body and subsequently uses the android camera to capture the lesion and stores it in the edge device. Then the user opens up the SkinAid application and identifies the visible symptoms. This is called the symptom analysis step. In the next step, the user uploads the lesion image and obtains the class of the lesion. The user also gets the contact of a dermatologist along with the class of the lesion. Figure 8a shows the presence of the lesion on the skin of the user and its successive magnification. Similarly, Fig. 8b shows the deployment results of SkinAid.

Table 6: Accuracies (in %) for various Deep Learning Models

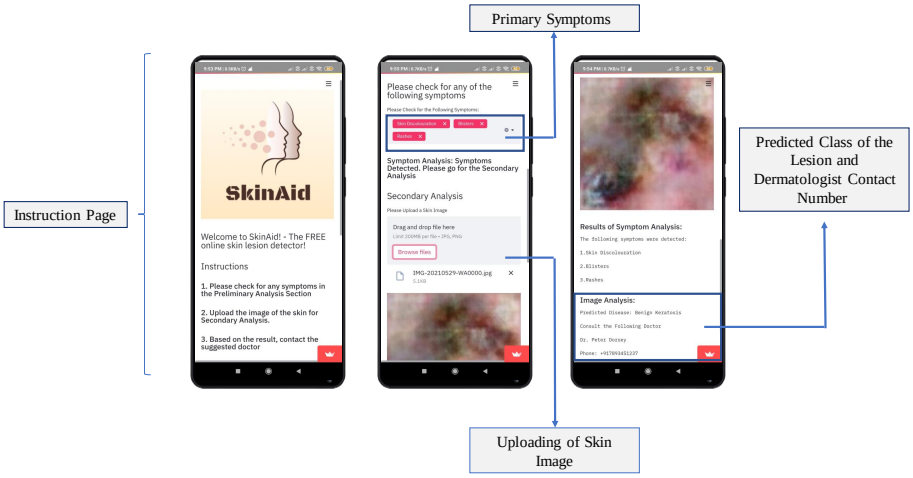
Sl.No	**Classifier Networks**	**Accuracy (%)**		
		Original Dataset	**Augmented Dataset**	
			No Preprocessing	**Preprocessed**
1	VGG-16	63.3	70.5	83.7
2	ResNet-50	68.1	71.9	87.3
3	ResNet-101	68.7	71.3	87.8
4	MobileNet-v2	66.4	69.8	89.6
5	Inception-v3	69.2	73.1	90.8
6	DenseNet-121	70.4	73.0	92.2

5.4 Performance comparison of different CNN Architectures

This section focuses upon the performance measures of the trained classifier on different Deep Learning Architectures. The different architectures are firstly trained on the orig-



(a) User Input on SkinAid Interface



(b) Detection and Classification of Skin Lesion

Fig. 8: Validation of SkinAid

inal dataset and their accuracy is recorded. They are then trained upon the augmented dataset involving both the conditions of non pre-processed images and pre-processed images and their accuracy is obtained. From Table 6, we can infer that DenseNet-121 has the highest accuracy amounting to 92.2%.

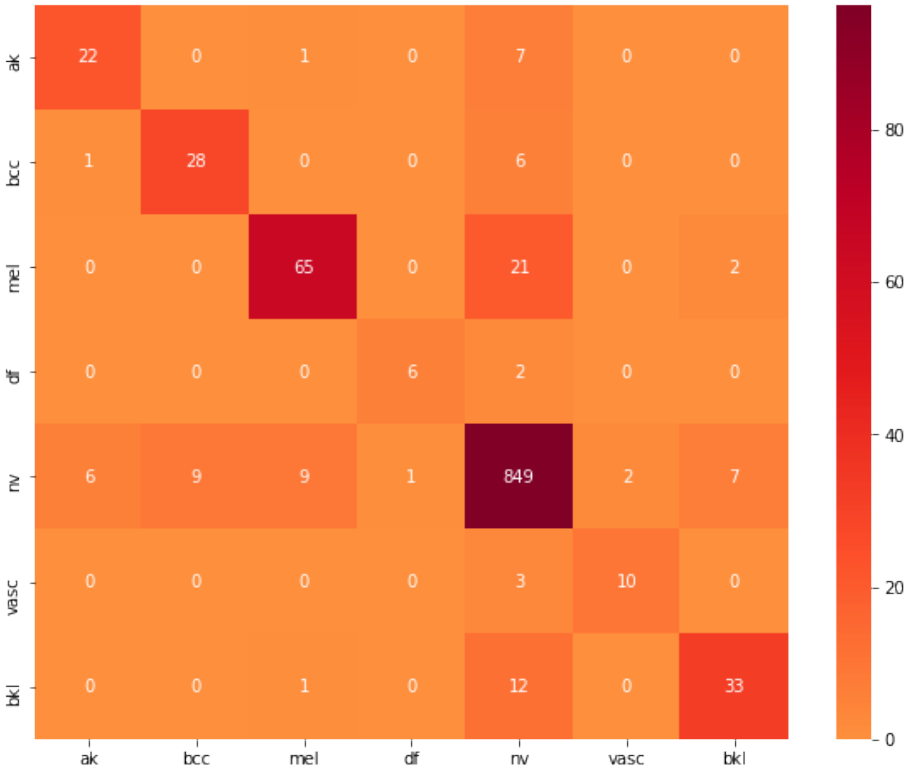


Fig. 9: Confusion Matrix

5.5 Confusion Matrix:

To justify the performance of our CNN model, we have also generated a confusion matrix that displays the relationship between the actual values and the predicted values of the model. Each element of the matrix is of the form $[l, m]$ where l represents the lth row and m represents the mth column of the matrix. If the value at $[l, m]$ where l and m are distinct is greater than 0, this implies that there has been a misclassification. If l, m are equal, it represents the true positive values. The following Figure. 9 shows the confusion matrix of the model

5.6 Precision, Recall and F1-Score

Precision is defined as the proportion of the true positives among all the identified positive values. Recall is defined as the proportion of the correctly identified actual positives. The F1-Score is a measure of the model's accuracy on the test set. Precisely, it can also stated that F1-Score is the harmonic mean of precision and recall. Let TP, TF, FP, FN denote the True Positives, True Negatives, False Positives and False Negatives predicted respectively. The Precision P is calculated by the following Equation 7:

$$P = TP/(TP + FP) \tag{7}$$

Similarly, Recall (R) can be calculated by the following equation 8:

$$R = TP/(TP + FN) \tag{8}$$

F1-Score is the harmonic mean of P and R. It can be denoted by the following equation 9:

$$F1 = 2 \times P \times R/(P + R) \tag{9}$$

The following Table. 7 depicts the precision, recall and F1-Score values of DenseNet-121:

Table 7: Precision, Recall and F1-Score of DenseNet-121

Class	Precision	Recall	F1-Score
Actinic Keratosis	0.75	0.73	0.739
Basil Cell Carcinoma	0.75	0.80	0.774
Melanoma	0.85	0.73	0.785
Dermatofibroma	0.85	0.75	0.796
Melanocytic Nevi	0.94	0.96	0.949
Vascular Lesions	0.83	0.76	0.792
Benign Keratosis	0.78	0.71	0.743

.

6 Conclusions and Future Work

In this paper, a smartphone aided skin lesion classification was formulated using GANs and deep CNNs. The app has been put into meticulous testing of skin lesions belonging to all the seven different classes and has obtained good results on all of them. The SkinAid app can be used by any user across the globe without the interference of internet. It empowers the users to detect potential lesions in an early phase, thus reducing the cost of treatment of the lesion aftermath.

In the subsequent aspects of the proposed solution, the SkinAid application can be improved to detect more classes of skin lesions with improved accuracy and efficient image data. Security and privacy issues in the Smart Health also needs research within the given constraints.

References

1. Albahar, M.A.: Skin lesion classification using convolutional neural network with novel regularizer. IEEE Access **7**, 38306–38313 (2019). https://doi.org/10.1109/ACCESS.2019.2906241
2. Arik, A., Gölcük, M., Karslıgil, E.M.: Deep learning based skin cancer diagnosis. In: Proc. 25th Signal Processing and Communications Applications Conference (SIU). pp. 1–4 (2017)
3. Arjovsky, M., Chintala, S., Bottou, L.: Wasserstein gan (2017). https://doi.org/1701.07875
4. Daghrir, J., Tlig, L., Bouchouicha, M., Sayadi, M.: Melanoma skin cancer detection using deep learning and classical machine learning techniques: A hybrid approach. In: Proc. 5th International Conference on Advanced Technologies for Signal and Image Processing (ATSIP). pp. 1–5 (2020)
5. Goswami, T., Dabhi, V., Prajapati, H.: Skin disease classification from image - a survey. pp. 599–605 (03 2020)
6. Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., Courville, A.: Improved training of wasserstein gans (2017)
7. Hossin, M.A., Rupom, F.F., Mahi, H.R., Sarker, A., Ahsan, F., Warech, S.: Melanoma skin cancer detection using deep learning and advanced regularizer. In: Proc. International Conference on Advanced Computer Science and Information Systems (ICACSIS). pp. 89–94 (2020)
8. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. In: Pereira, F., Burges, C.J.C., Bottou, L., Weinberger, K.Q. (eds.) Proc. Advances in Neural Information Processing Systems. vol. 25. Curran Associates, Inc. (2012)
9. Li, X., Wu, J., Jiang, H., Chen, E.Z., Dong, X., Rong, R.: Skin lesion classification via combining deep learning features and clinical criteria representations. bioRxiv (2018). https://doi.org/10.1101/382010, `https://www.biorxiv.org/content/early/2018/08/01/382010`
10. Lin, R., Wong, E.: Morphological operations on images represented by quadtrees. In: Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing Conference Proceedings. vol. 4, pp. 2203–2206 vol. 4 (1996)
11. Mahbod, A., Schaefer, G., Wang, C., Ecker, R., Ellinge, I.: Skin lesion classification using hybrid deep neural networks. In: Proc. ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). pp. 1229–1233 (2019). https://doi.org/10.1109/ICASSP.2019.8683352
12. Manoorkar, P.B., Kamat, D.K., Patil, P.M.: Analysis and classification of human skin diseases. In: Proc. International Conference on Automatic Control and Dynamic Optimization Techniques (ICACDOT). pp. 1067–1071 (2016)
13. Mitra, A., Mohanty, S.P., Corcoran, P., Kougianos, E.: A machine learning based approach for deepfake detection in social media through key video frame extraction. SN Comput. Sci. **2**(2), 98 (2021). https://doi.org/10.1007/s42979-021-00495-x
14. Rachakonda, L., Bapatla, A.K., Mohanty, S.P., Kougianos, E.: SaYoPillow: Blockchain-integrated privacy-assured iomt framework for stress management considering sleeping habits. IEEE Transactions on Consumer Electronics **67**(1), 20–29 (2021). https://doi.org/10.1109/TCE.2020.3043683
15. Rachakonda, L., Mohanty, S.P., Kougianos, E.: iLog: An intelligent device for automatic food intake monitoring and stress detection in the iomt. IEEE Transactions on Consumer Electronics **66**(2), 115–124 (2020). https://doi.org/10.1109/TCE.2020.2976006
16. Rezaoana, N., Hossain, M.S., Andersson, K.: Detection and classification of skin cancer by using a parallel cnn model. In: Proc. IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE). pp. 380–386 (2020)

17. Romero Lopez, A., Giro-i Nieto, X., Burdick, J., Marques, O.: Skin lesion classification from dermoscopic images using deep learning techniques. In: Proc. 13th IASTED International Conference on Biomedical Engineering (BioMed). pp. 49–54 (2017). https://doi.org/10.2316/P.2017.852-053
18. Shah, V., Autee, P., Sonawane, P.: Detection of melanoma from skin lesion images using deep learning techniques. In: Proc. International Conference on Data Science and Engineering (ICDSE). pp. 1–8 (2020). https://doi.org/10.1109/ICDSE50459.2020.9310131
19. Tschandl, P., Rosendahl, C., Kittler, H.: The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Scientific Data **5**(1) (Aug 2018). https://doi.org/10.1038/sdata.2018.161, `http://dx.doi.org/10.1038/sdata.2018.161`
20. Vidya, M., Karki, M.V.: Skin cancer detection using machine learning techniques. In: Proc. IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT). pp. 1–5 (2020)